

1 Important instructions

1. No documents are allowed and no calculator nor any electronic device.
2. If your paper is not clearly legible or not sufficiently well presented, points may be deducted.
3. Exercises are independent. You can answer questions in any order, provided that your answers are clearly numbered.
4. Read instructions for each exercise

2 Map-Reduce: Communication Network (5 pts)

- You should write the code in Spark to create the given RDD in each question.
- If you build intermediate RDD to answer a question, explain what is this RDD and its format.
- The Spark functions you can use to solve the exercises are: `map`, `flatMap`, `reduceByKey`, `filter` and `union`. Other functions are not allowed.

We consider a communication network. Each user is identified by a number UID. We assume each user can send messages to other users. Logs of these messages are stored in a RDD `Message` of tuples (MID, UID1, UID2, D, M, Y, SIZE). All elements are integers. MID is the message ID of a message send by user UID1 to user UID2 on the date (Day, Month, Year); SIZE is the size of the message (in bytes).

As an example, we assume that the `Message` RDD is stored without duplication on two nodes N1 and N2 as follow: (of course, your code should work on any RDD, not only on this example)

On N1:

MID	UID1	UID2	D	M	Y	SIZE
1	1	2	12	07	2025	105
2	2	3	11	08	2025	25
3	1	3	14	07	2025	305
4	2	3	21	09	2024	78
5	2	3	03	10	2024	5

And on N2:

MID	UID1	UID2	D	M	Y	SIZE
6	1	3	14	05	2024	2
7	1	2	12	07	2025	105
8	1	3	11	08	2025	25
9	3	2	14	07	2025	305
10	2	3	21	09	2024	78

Question 1 : Create a `MsgTot` RDD containing tuples (UID, #MSG_R) where #MSG_R is the number of messages received by each user UID.

Question 2 : Assume in this question that there is no pre-reduce optimization on nodes. Draw the computation tree(s) of the reduce operation for the previous question on the example RDD. How many tuples are send on the network during the shuffle step? In the tree, identify each input tuple with its MID (even if you do not keep the MID in your computation).

Question 3 : Same question, but we assume pre-reduce optimization is used.

Question 4 : For each user and each month/year, we want to compute the number of important recipients. For a user A, user B is an important recipient if A send more than 1000 bytes of message (in total) in the given month/year to user B. Create a `Imp_R` RDD containing tuples (UID, M, Y, NBIR) where NBIR is the number of important recipients for user UID in month/year given by M/Y.

next on back

3 Recommendation Systems (2 pts)

The following table represents a part of a utility matrix R giving the ratings of users (in rows) on different items (columns): r_{ij} is the rating of user i for item j .

If an entry of the matrix is empty, it means the user did not evaluate this item.

R	I1	I2	I3	I4	I5	I6	I7
User 1		2	1		5		4
User 2		3		2		3	1
User 3	4	1			1	3	5
User 4		3		1	4		
...							

The following table gives the similarity (between 0 and 1) between items (the matrix is symmetric, so only the upper triangle is given).

Sim	I1	I2	I3	I4	I5	I6	I7
I1	1	0.1	0.3	0.6	0.4	0.5	0.8
I2		1	0.7	0.2	0.5	0.2	0.5
I3			1	0.5	0.6	0.4	0.2
I4				1	0.7	0.2	0.6
I5					1	0.4	0.1
I6						1	0.2
I7							1

We want to compute estimation with basic item-item collaborative filtering using 2 neighbors for the estimation.

Question 5 : Given these similarities and the ratings, compute an estimate of the rating r_{11} of user 1 for item 1. You can give the result as a ratio. Explain the steps of this estimation (which items will be used and why).

Question 6 : We now want an estimate of the rating r_{23} of user 2 for item 3. Explain which items will be used in the estimation and why. (You do not need to compute the estimate.)

4 Locality Sensitive Hashing (3 pts)

In the following matrix, each column represent a set (or document) and each row an element (or shingle). There is a '1' in row i and column j if element i belongs to set j .

	A	B	C	h_1	h_2
0	1	0	1	5	4
1	1	1	1	6	3
2	1	1	0	3	5
3	1	1	0	4	6
4	0	0	0	0	1
5	0	1	0	2	2
6	0	0	1	1	0

Question 7 : Give the formulas and the value of the Jaccard similarities between sets A and B, B and C, and A and C.

Question 8 : We use the hash functions h_1 and h_2 in the table above to define permutations of rows/elements (for instance, row 0 means $h_1(0) = 5$, $h_2(0) = 4$). Using the row by row (optimized) algorithm, compute the minhash of A, B and C according to permutations h_1 and h_2 (give the table of signatures after each row).

Recap of Row by row optimized algorithm:

For each set S and for each hash function H:

Initialize signature(S, H) to infinity.

For each row R in increasing order

For each set S and for each hash function H:

if $S(R)=1$ then $\text{signature}(S, H) = \min(\text{signature}(S, H), H(R))$

$S(R)$ denotes the value of set S in row R. At the end of the algorithm, $\text{signature}(S, H)$ is the minhash value of set S for function H.